Alejandro Soler Gonzálvez

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Physicist (Master in Space Systems, **GPA 8.5/10**, thesis 9.5/10) moving into **quantitative research & trading**. Built a cross-venue data-capture system in **Rust** to study how two prediction markets price the same Bitcoin hour; validated the effect on 1,350 real settlements and published it as an SSRN research note. **Available now for internships**; planning an M.Sc. in quantitative finance from autumn 2027.

Education

Master in Space Systems (Máster de Formación Permanente, 60 ECTS) — Universidad Europea 2025 – 2026
GPA 8.5/10

- **Master's Thesis (9.5/10):** fuel-optimal guidance for autonomous rocket landing — posed the final 2 km of powered descent as a **convex program (SOCP)** via lossless convexification, optimized the flight time over a sweep of convex solves and flew the solution in a self-built **3-DoF simulator**. Reproduced a published Mars-lander benchmark (75 s, 373.76 kg of fuel, bang-bang thrust); same convex-guidance family as SpaceX booster landings. Interactive demo.

B.Sc. in Physics — Universidad Complutense de Madrid (bilingual Spanish/English program) 2021 – 2025

- **Bachelor's Thesis (9.2/10):** multi-agent **Bayesian search** — coordinated agents locate the maximum of an unknown field via **Gaussian-process (Kriging) regression**, balancing exploration vs. exploitation under uncertainty. Selected for Docta Complutense (UCM's distinguished-thesis repository). Interactive demo.
- Coursework: mathematical methods, statistical physics, scientific computing, probability & statistics; quantum-computing program (3 algorithms implemented in **Qiskit**).

Experience & Projects

Quantitative Research — Cross-Venue BTC Oracle Basis (Polymarket / Kalshi) Feb – May 2026

- Discovered a **systematic, signed cross-venue mispricing** in hourly Bitcoin prediction markets: the two venues settle on **different oracles** (Binance USDT vs. CF Benchmarks USD), so their implied strikes disagree by the **USDT/USD stablecoin basis** — signed and forecastable before each hour resolves.
- Validated the mechanism on **1,350 real Kalshi settlements**: an offset-aware entry rule **removes the double-loss tail** (4.6% $\rightarrow$ 0%) and isolates the $\sim$21% regime where both legs can win; characterized the two-sided cost distribution from live orderbook capture. Published it as a research note on SSRN, with an interactive companion.
- Built the 24/7 data-capture engine behind the study in **Rust** (chosen over Python for latency) on **AWS eu-west-1 (Ireland)**: low-latency access to both venues, live order-book capture and settlement tracking. Designed it and **rebuilt the whole system on my own**.

Flight Dynamics Intern — PLD Space (launch-vehicle company, Elche, Spain) Mar – Apr & Jun – Aug 2025

- Built **MATLAB** tooling that automated recurring flight-dynamics analyses; reworked the **numerical surface-discretization scheme** behind the vehicle's aerodynamic force integration, improving its accuracy.

Co-founder & Sole Developer — Bolbee, AI marketing platform for SMEs 2025 – 2026

- Built the full product end-to-end (Next.js, Supabase/SQL, Stripe, multi-model AI pipeline) — architecture, data model, infrastructure, integrations. Secured an investor, reached private beta; chose not to launch after the investor left.

Team Lead — Exoplanet Detection with ML, NASA Space Apps Challenge (Madrid) 2025

- Led a 3-person team building a neural-network exoplanet classifier on NASA telescope data (**Python**): owned data cleaning and feature selection, defined the train/test methodology and evaluation (confusion matrix); delivered a working model in 48 h.

Technical Skills

Programming: Python (NumPy, pandas — research analysis & backtesting for the 1,350-settlement cross-venue study; data/eval pipeline for the NASA project), Rust (low-latency data-capture engine), MATLAB (flight dynamics, simulators), SQL, TypeScript.
Quantitative: convex optimization, probability & statistics, market microstructure, Gaussian processes, Monte Carlo methods, control & estimation theory, numerical optimization.
Tools: Git, AWS, REST APIs.

Languages

Spanish (native) | English (C1, Cambridge certified) | Catalan (C1) | French (A2)

Honors & Additional

- Merit scholarship: full first-year university tuition (ranked 1st in graduating high-school class).